

Correcting

Exercise nº 3

On the *Prelude no. 4 in E minor, Op. 28*, by Frédéric Chopin (1839) — public domain.

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What you will do: open in the Player a MIDI file of the Prelude that has **wrong notes on purpose**, find them by ear with the score beside you, and fix them with the **copyist**. At the end you will save your corrected version in three ways: the whole work, one hand only, and a passage of bars.

How long: about thirty minutes. **What you need:** the files of this lesson (`Preludio Nº 4 Op28 - E min (para corregir).mid` and the **PDF** of the score), the **COSMIC KEYBOARD** open for the sound, and Exercise nº 2 done.

The copyist's moves

In Pythagorean tuning every note has its exact place, and the copyist moves it by **true intervals**, not by keys:

±T **Tone**. Up or down a whole tone (C → D).

±L **Limma**, the diatonic semitone: **the letter changes** (E → F, C → B).

±A **Apotome**, the chromatic semitone: **the letter stays**, the accidental changes (F → F♯).

±K **Comma** (23.46 cents): the **comma neighbour** (A♭ ↔ G♯). They are two different notes.

A tone = a limma + an apotome. That is why «up a semitone» has **two paths** that do not sound the same: with L the letter changes, with A the accidental changes.

PART ONE — FIND AND FIX

IN THE PICTURES: the orange circles show where to look or click.

1 • Load the MIDI file into the PLAYER

Open the Player from the Cosmic Keyboard and drop the file

`Preludio Nº 4 Op28 - E min (para corregir).mid` on the drop zone. Listen to all of it with

, with the PDF score in front of you. Does anything sound odd?



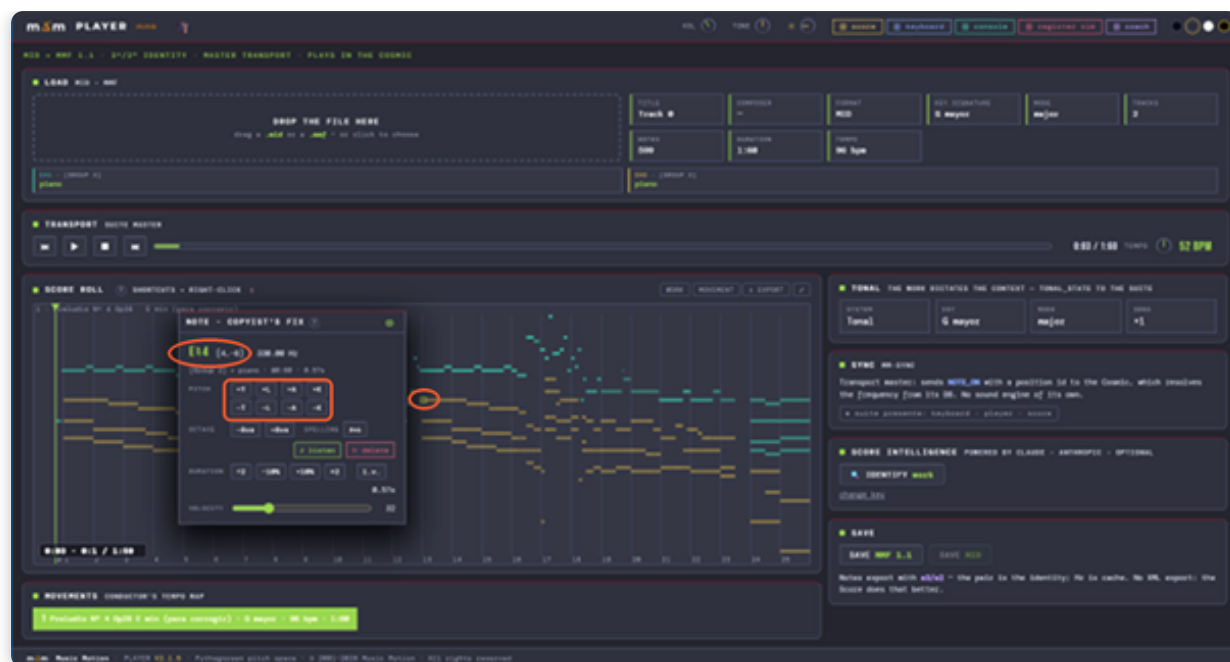
2 • Move around the Score Roll

- **Mouse wheel:** moves up and down through the pitches.
- **Shift + wheel:** moves forward or back in time.
- **Ctrl + wheel:** zooms in or out, around the cursor.
- **Click and drag:** moves the view when zoomed in.
- **Double-click an empty spot:** back to the whole work.

Zoom first. With the whole work in view, the wheel and the drag have nowhere to go: zoom in with **Ctrl + wheel** and from then on everything moves.

Below the roll, the **bar ruler** numbers every barline as in the score (the upbeat is bar 0), and the clock always tells you **bar:beat**. The green track is the **right hand** and the orange one the **left hand**.

One click on a note opens the **copyist**: its name, its exact frequency and the buttons to move it. With **listen** you hear it on its own.



3 • The first mistake

One note of the **melody** is higher than the score says. Find it by comparing with the PDF.

► Hint

► Solution

4 • The second mistake

Another note of the melody **lost its accidental**. Look for a missing sharp.

► Hint

► Solution

Check: listen to the whole Prelude again. If the two corrected notes no longer bother you, it is right. The notes you touched are marked as *edited* in the copyist.

5 • Discover the comma

This step fixes nothing: it is for **hearing** something a tempered keyboard does not let you hear. In **bar 3, beat 4**, the left hand has a **G[#]3**. Open it in the copyist.

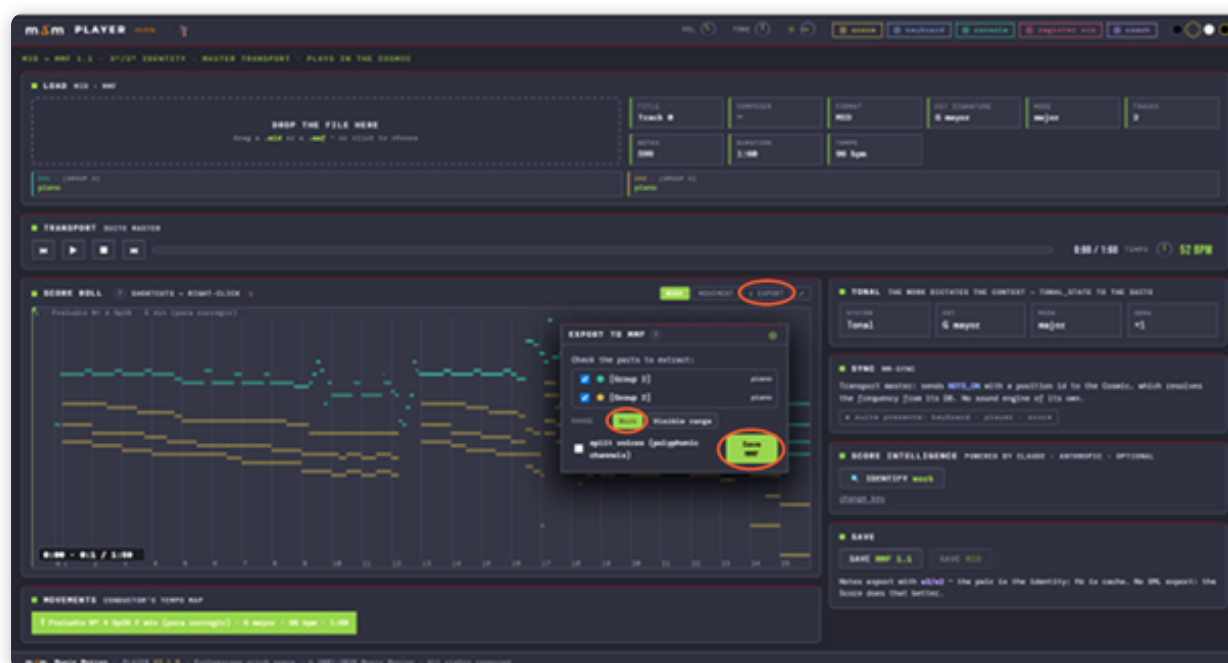
1. Listen to it with listen .
2. Press . Now it says **Ab3**: the letter went up from G to A, but the sound went **down** a comma. Listen to it.
3. Press to go back to G[#].

Which of the two pulls towards the A above? G♯ is the leading note: it is higher and wants to rise. A♭ is another note, a comma lower, that wants to fall. On a piano they are the same key; here they are not the same note. Leave it on **G♯**, which is what the harmony asks for.

PART TWO — SAVE YOUR VERSION

6 • The whole work

1. In the Score Roll header, button **↓ export** → **Export by instrument...**
2. Leave **both tracks** checked and the range on **Work**.
3. **Save MMF**. The file downloads with the name of the tracks.



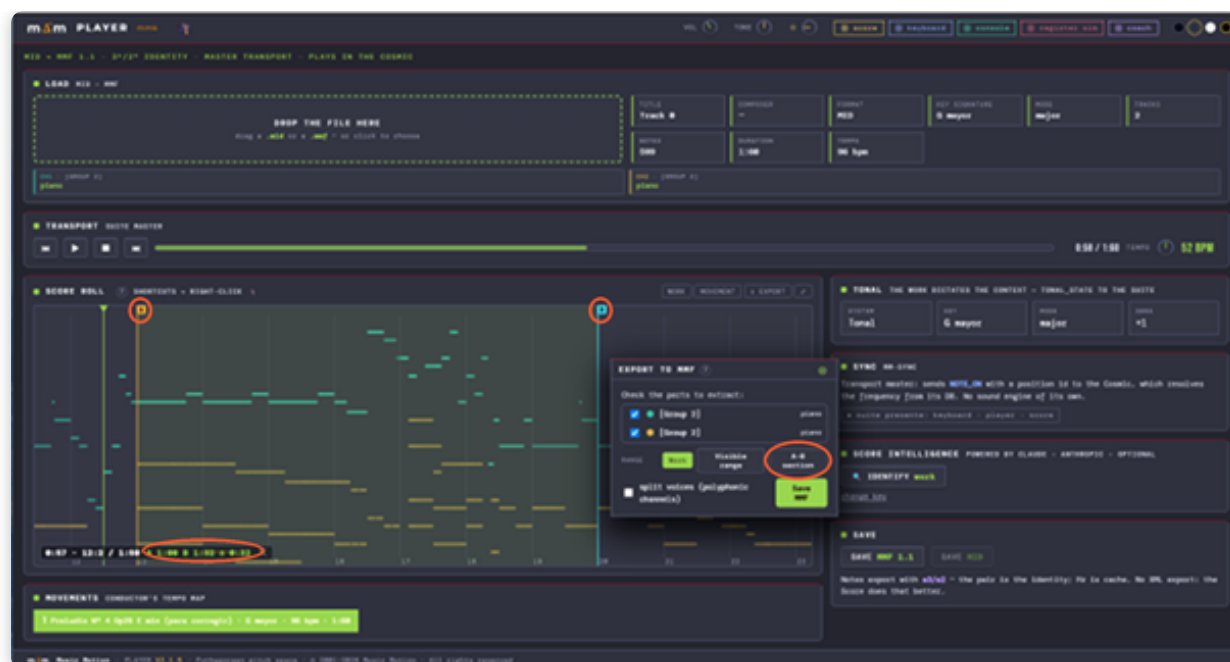
7 • One hand only

The same, but **uncheck one of the two tracks**: only the green one for the right hand, or only the orange one for the left. Save both, one after the other. They are for practising each hand on its own, or for the Score to open them as separate parts.

8 • A passage: bars 13 to 19

1. Find **bar 13** on the ruler. Right-click right on its barline → **Set A here**.
2. Find the barline of **bar 20** (the end of 19). Right-click → **Set B here**. The section is shaded between A and B.
3. **↓ export** → **Export by instrument...** Now the range has one more option: **A-B section**. Choose it, check the tracks you want and **Save MMF**.

TIP: zoom in with **Ctrl + wheel** before setting A and B: the click then lands right on the barline. If it slipped, drag the little A or B flag to its place.



Check: your downloads folder now has several `.mmf` files: the corrected work, each hand and the passage. Open them in the Score (`MENU FILES` → `IMPORT MMF`) and see that each one holds what you chose.

Congratulations: you corrected and saved your own copy

In **Exercise nº 4** you open the corrected work in the Score and finish writing it: dynamics, slurs, beams and the details a MIDI file never brings.